- $G_1 = \{\pm 1, \pm i\} \times \{\mathbb{1}, X, Y, Z\}$

- $G_n = = \{\pm 1, \pm i\} \times \{\mathbb{1}, X, Y, Z\}^{\otimes n}$ PAULI GROUP $\left(|G_n| = 4^n \cdot 4\right)$

  ↑ phases necessary to be a group.

PROPERTIES OF G_n (@)

A) • $P_A, P_B \in G_n \Rightarrow P_A P_B = P_B P_A$ or $P_A P_B = -P_B P_A$. (commute or anticommute)

B) • $P \in G_n \Rightarrow P$ is unitary $PP^+ = \mathbb{1}$.

C) • $P \in G_n \setminus \{\pm \mathbb{1}, \pm i\mathbb{1}\} \Rightarrow \mathrm{tr}(P) = 0$

D) • $P \in G_n$: $P = \pm 1 \cdot \hat{P}$, $\hat{P} \in \{\mathbb{1}, X, Y, Z\}^{\otimes n} \Rightarrow P^2 = \mathbb{1}$, $P = P^+$, eigenv. $= \pm 1$

$P \in G_n$: $P = \pm i \cdot \hat{P}$, $\hat{P} \in \{\mathbb{1}, X, Y, Z\}^{\otimes n} \Rightarrow P^2 = -\mathbb{1}$, $P^+ = -P$, eigenval $= \pm i$.

E) • $P \in G_n \Rightarrow |\langle\psi| P |\psi\rangle| \leq 1$

(WRITE $|\psi\rangle$ in the eigendecomposition)

F) • $P \in G_n \Rightarrow |\langle\psi| P |\psi\rangle| = 1 \Rightarrow |\psi\rangle$ eigenstate of P.

• DEF (1):
Given $S \subseteq G_n$ subgroup, we define V_S the set of n-qubit state $|\psi\rangle$ such that $S_i |\psi\rangle = +|\psi\rangle \quad \forall S_i \in S$.

• OBS (2): V_S is a vector space.

• DEF (3) V_S is called "vector space stabilized by S".

• DEF (4): Given $S \subseteq G_n$ subgroup, we indicate $S = \langle S_1, \dots, S_\ell \rangle$ if $\mathrm{Gen}(S) := \{S_1, \dots, S_\ell\}$ are generators for S (i.e. $\forall P \in S$, P can be written as product of elements taken from $\{S_1, \dots, S_\ell\}$.).

EXAMPLE (3) $S = \langle \mathbb{1}, X_1 X_2 \rangle = \{\mathbb{1}, X_1 X_2\}$

- $\mathbb{1}|\psi\rangle = |\psi\rangle$
- $X_1 X_2 |\psi\rangle = |\psi\rangle$

$\Leftrightarrow V_S = \mathrm{Span}\left(\left\{ \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \right\} \right)$

- OBS (6): $\dim(V_S) > 0 \Rightarrow$ A) $-\mathbb{1} \notin S$

B) $S_i, S_j \in S \Rightarrow [S_i, S_j] = 0$

PROOF:

(A) · Assume $-\mathbb{1} \in S$.

$|\psi\rangle \in V_S \Rightarrow -\mathbb{1}|\psi\rangle = |\psi\rangle \Rightarrow |\psi\rangle = 0 \Rightarrow \dim(V_S) = 0 \Rightarrow$ ABSURD.

(B) · Assume that $\exists\, S_i, S_j \in S : [S_i, S_j] \neq 0 \Rightarrow$ (S_i, S_j Pauli's or commute or anticommute.)

· $|\psi\rangle \in V_S \Rightarrow |\psi\rangle = S_i S_j |\psi\rangle = -S_j S_i |\psi\rangle$

$= -|\psi\rangle \Rightarrow |\psi\rangle = 0$

$\Rightarrow \dim(V_S) = 0 \Rightarrow$ ABSURD.

- COR (7): $\dim(V_S) > 0 \Rightarrow$ A) $-\mathbb{1} \notin S$

B) $S_i, S_j \in \mathrm{Gen}(S) \Rightarrow [S_i, S_j] = 0$

- FACT (8): We will prove that also the converse is true.

OBS (9) $S = \langle S_1, \dots, S_L \rangle$. $[S_i, S_j] = 0$ with $S_i, S_j \in \mathrm{Gen}(S) \Leftrightarrow S$ is abelian.

PROOF:

· ($\Leftarrow$) ok. · ($\Rightarrow$) $\forall\, g_1, g_2 \in S \Rightarrow g_1$ and g_2 can be written as product of $\mathrm{Gen}(S)$ elements $\Rightarrow$ II.

- OBS (10): $-\mathbb{1} \notin S \Rightarrow \left(S_i, S_j \in \mathrm{Gen}(S) \Rightarrow [S_i, S_j] = 0 \right)$

PROOF:

Suppose that $\exists\, g$ and $h \in \mathrm{Gen}(S) : hg = -gh$.

If $h^2 = -\mathbb{1}$ or $g^2 = -\mathbb{1} \Rightarrow$ ok. If $h^2 = \mathbb{1}$, $g^2 = \mathbb{1} \Rightarrow (gh)^2 = ghgh = -g^2h^2 = -\mathbb{1} \in S$

$\Rightarrow$ ABSURD.

OBS: (2.1) $-\mathbb{1} \notin S \Rightarrow \forall P \in S \quad P = \pm Q$ with $Q \in \{\mathbb{1}, X, Y, Z\}^{\otimes n}$

$(P \neq \pm i Q)$

$\Rightarrow \forall P \in S, \quad P^2 = \mathbb{1}.$

PROOF:

If $P = (\pm i Q) \in S \Rightarrow P^2 = -\mathbb{1} \in S.$

• COR. (2.2) $\dim(V_S) > 0 \Rightarrow -\mathbb{1} \notin S \Rightarrow$ A) • S is abelian ($\mathrm{Gen}(S)$ also).

B) • $\forall P \in S, \quad P = \pm Q, \quad P = P^+$ ($Q \in \{\mathbb{1}, X, Y, Z\}^{\otimes n}$)

C) • $\forall P \in S, \quad P^2 = \mathbb{1}$, eigenvalues: ± 1.

• EXAMPLE (2.3) • $S = \langle \mathbb{1}, iP \rangle \Rightarrow V_S = \{0\}$

$iP \in S \Rightarrow (iP)^2 = -\mathbb{1}$

• $S = \langle \mathbb{1}, X_1, Y_1 \rangle \Rightarrow V_S = \{0\}$

$X_1 Y_1 = -Y_1 X_1$

• TH (2.4): • The projector on V_S is given by $P_S := \frac{1}{|S|} \sum_{g \in S} g$

PROOF:

• If $-\mathbb{1} \in S \Rightarrow$ if $g \in S \Rightarrow -\mathbb{1} g = -g \in S \Rightarrow P_S = 0$. ok!

• If $-\mathbb{1} \notin S \Rightarrow$ • $P_S = P_S^+$ (2.2), $P_S^2 = P_S \Rightarrow$ PROJECTOR.

$$P_S^2 = \left(\frac{1}{|S|}\sum_{g\in S} g\right)\left(\frac{1}{|S|}\sum_{g'\in S} g'\right) = \frac{1}{|S|^2}\sum_{g}\underbrace{\left(\sum_{g'} gg'\right)}_{\left(\sum_{g''} g''\right)} = \frac{1}{|S|^2}\left(\sum_{g} 1\right)\cdot\left(\sum_{g'} g'\right) = P_S$$

$\sum_g 1 = |S|$

- $P_S|\psi\rangle = |\psi\rangle \iff |\psi\rangle \in V_S$.

SUBPROOF:

$$P_S|\psi\rangle = |\psi\rangle \Rightarrow 1 = \langle\psi|P_S|\psi\rangle = \frac{1}{|S|}\sum_g \langle\psi|g|\psi\rangle \leq \frac{1}{|S|}\sum_g |\langle\psi|g|\psi\rangle| \overset{\text{a.f.}}{\leq} \frac{1}{|S|}\sum_g 1 = 1$$

$$\Rightarrow \langle\psi|g|\psi\rangle = 1 \overset{\text{O.F.}}{\Rightarrow} g|\psi\rangle = +|\psi\rangle \ \forall g\in S \Rightarrow |\psi\rangle \in V_S.$$

- **TH (5)**

- $S = \langle S_1, \dots, S_\ell\rangle$ where $\text{Gen}(S) = \{S_1, \dots, S_\ell\}$ are indipendent generators.

If $-I \in S \Rightarrow \dim(V_S) = 0$

If $-I \notin S \Rightarrow$
- $P_S := \frac{1}{2^\ell}\sum_{g\in S} g = \prod_{k=1}^{\ell}\left(\frac{1+S_k}{2}\right)$
- $\dim(V_S) = 2^{n-\ell}$

PROOF:

- If $-\mathbb{1} \in S \Rightarrow$ if $g\in S \Rightarrow -\mathbb{1}g = -g \in S \Rightarrow P_S = 0$. OK!
- If $-\mathbb{1} \notin S \Rightarrow P_S := \frac{1}{|S|}\sum_{g\in S} g \overset{\uparrow}{=} \frac{1}{|S|}\prod_{k=1}^{\ell}(\mathbb{1}+g_k) =$

  $\uparrow$ generators

  · COR (4.9) $\Rightarrow$ S is abelian and $g^2 = \mathbb{1}$, $\forall g\in S$.

$$= \frac{1}{(|S| = 2^\ell)\,2^\ell}\prod_{k=1}^{\ell}\left(\frac{\mathbb{1}+S_k}{2}\right)$$

- $\dim(V_S) = \text{Tr}(P_S) = \frac{1}{2^\ell}\text{Tr}\left(\sum_{g\in S} g\right) = \frac{2^n}{2^\ell} = 2^{n-\ell}$.

  $\uparrow$ $\text{Tr}(g) = 0 \ \forall g \neq \mathbb{1}$

TH (26) • $S = \langle S_1, \dots, S_n \rangle$ with $S_1, \dots, S_n$ independent generators such that $-\mathbb{1} \notin S$.

$\Rightarrow \dim(V_S) = 1$

PROOF: Follows by (25).

FACT (27)

• Given $Gen(S) = \{S_1, \dots, S_\ell\}$, how do we verify that $-\mathbb{1} \notin S$?

- If $S_1, \dots, S_\ell$ commute
- $S_1, \dots, S_\ell$ are independent removing phase factors.
- $S_i^2 = \mathbb{1} \quad \forall S_i \in Gen(S)$ ($S_i = \pm P$ with $P \in \{\mathbb{1}, X, Y, Z\}^{\otimes n}$)
- $S_i \neq -\mathbb{1} \quad \forall S_i \in Gen(S)$

$\Rightarrow \quad -\mathbb{1} \notin S$

PROOF:

• $-\mathbb{1} \in S \Rightarrow -\mathbb{1} = S_1^{x_1} \cdots S_\ell^{x_\ell}$ for some $x_1, \dots, x_\ell \Rightarrow -\mathbb{1} = \tilde{S}_1 \cdots \tilde{S}_m$

(considering only $S_i^{x_i}$: $x_i \neq 0$.)

• Here $m > \ell$, since if $m = 1 \Rightarrow -\mathbb{1} = \tilde{S}_1 \in Gen(S)$, $\Rightarrow$ ABSURD.

• $\tilde{S}_m = -\tilde{S}_1 \cdots \tilde{S}_{m-1} \Rightarrow S_m$ and $\tilde{S}_1, \dots, \tilde{S}_{m-1}$ are NOT independent removing phase factors. $\Rightarrow$ ABSURD.

TH (18)

A) • If $S_1, \dots, S_\ell$ commute

B) • $S_1, \dots, S_\ell$ are indipendent removing phase factors. $\Rightarrow \dim\left(V_{S=\langle S_1,\dots,S_\ell\rangle}\right) = 2^{n-\ell}$

C) • $S_i^2 = \mathbb{1} \quad \forall S_i \in Gen(S)$
($S_i = \pm P$ with $P \in \{\mathbb{1}, X, Y, Z\}^{\otimes n}$)

D) • $S_i \neq -\mathbb{1} \quad \forall S_i \in Gen(S)$

PROOF:

(14) + (15) .

FACT (19)

If we have $S_1, \dots, S_n$ and we have to verify conditions of TH (18), C) and D) can be done efficiently. But what about A) and B)?

We need the so-called CHECK-REPRESENTATION tool.

DEF (20) (CHECK-REPRESENTATION)

• We define a $2n$-dim. vector representation $R(P)$ of a Pauli $P = \pm P_1 \otimes P_2 \otimes \dots \otimes P_n$

$$P_i = \mathbb{1} \iff \left(R(P)\right)_i = 0 \;,\; \left(R(P)\right)_{n+i} = 0$$

$$P_i = X \iff \left(R(P)\right)_i = 1 \;,\; \left(R(P)\right)_{n+i} = 0$$

$$P_i = Z \iff \left(R(P)\right)_i = 0 \;,\; \left(R(P)\right)_{n+i} = 1$$

$$P_i = Y \iff \left(R(P)\right)_i = 1 \;,\; \left(R(P)\right)_{n+i} = 1$$

e.g. $P = X \otimes 2 \otimes \mathbb{1} \otimes Y \Rightarrow R(P) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ \hline 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \} x \\ \\ \\ \\ \} 2 \end{matrix}$

$\quad = 2X$

· We don't consider phases in this representation.

FACT (2.1) $P^{(A)} P^{(B)} = \pm P^{(C)} \Rightarrow \left(R(P^{(A)}) + R(P^{(B)}) = R(P^{(C)}) \right) \mod(2)$.

PROOF:

$$P^{(A)} P^{(B)} = \left(P_1^{(A)} \otimes \cdots \otimes P_n^{(A)}\right)\left(P_1^{(B)} \otimes \cdots \otimes P_n^{(B)}\right) = P_1^{(A)} P_1^{(B)} \otimes \cdots \otimes P_n^{(A)} P_n^{(B)}$$

We can verify it case by case.

e.g. $P_5^{(A)} = X$, $P_5^{(B)} = Y$ $\Rightarrow XY = i Z = P_5^{(C)}$

$$\underset{\shortparallel}{(R(X))_5} + \underset{\shortparallel}{(R(Y))_5} = \underset{\shortparallel}{(R(XY))_5} = (R(Z))_5$$
$$1 \qquad 1 \qquad = \qquad 0 \qquad \text{ok!}$$

$$\underset{\shortparallel}{(R(X))_{5+n}} + \underset{\shortparallel}{(R(Y))_{5+n}} = \underset{\shortparallel}{(R(XY))_{5+n}} = (R(Z))_{5+n}$$
$$0 \qquad 1 \qquad = \qquad 1 \qquad \text{ok!}$$

· (2.2) More generally:

$Q = \pm S_n^{x_n} \cdots S_1^{x_1}$ with $x_1, \ldots, x_n \in \{0,1\} \Rightarrow R(Q) = x_1 R(S_1) + \cdots + x_n R(S_n)$

$$R(Q) = x_1 R(S_1) + \cdots + x_n R(S_n) = \underbrace{\left(\begin{array}{c|c|c|c} R(S_1) & R(S_2) & \cdots & R(S_n) \end{array}\right)}_{\text{2n} \times \text{n matrix}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} =: \underset{=}{R_S} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$\underset{=}{R_S}$

(23) To check if $\{S_1, \dots, S_\ell\}$ are independent without phases (check-indep.)
we need to see if $\left(R(S_1)\,|\,\dots\,|\,R(S_\ell)\right)\begin{pmatrix} x_1 \\ \vdots \\ x_\ell \end{pmatrix} = 0$
admit only $x_1 = \dots = x_\ell = 0$ as solution.
These can be solved efficiently in $O(\ell^3)$ time
using Gaussian elimination

(24) How to check if $\{S_1, \dots, S_\ell\}$ commute each other?
I need to check if $[S_i, S_j] = 0 \quad \forall i \neq j$.

$\lesssim \dfrac{n(n-1)}{2}$.

- Checking if $S_i S_j = S_j S_i$ $(i \neq j)$ can be done also using the check-regress.

FACT (25) $S_i S_j = S_j S_i \iff \left(R(S_i)\right)^t \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} R(S_j) = 0$

$2n \updownarrow$; $\mathbb{1}$: $\updownarrow n$, $\leftrightarrow n$; 0: $\updownarrow n$, $\leftrightarrow n$

PROOF:

- $\sum_{K=1}^{n} \left(R(S_i)\right)_K \left(R(S_j)\right)_{n+K} + \sum_{K=1}^{n} \left(R(S_i)\right)_{K+n} \left(R(S_j)\right)_K = 0$

- $\sum_{K=1}^{n} \left(\left(R(S_i)\right)_K \left(R(S_j)\right)_{n+K} + \left(R(S_i)\right)_{K+n} \left(R(S_j)\right)_K \right) = 0$

- $S_i = P_1^i \otimes \dots \otimes P_n^i$, $S_j = P_1^j \otimes \dots \otimes P_n^j$

- $S_i S_j = P_1^i P_1^j \otimes \dots \otimes P_n^i P_n^j$
- $S_j S_i = P_1^j P_1^i \otimes \dots \otimes P_n^j P_n^i$
- $P_k^i P_k^j = P_k^j P_k^i$ or $P_k^i P_k^j = -P_k^j P_k^i$

This situation should happen an even number of times for commute.

I have to verify that:

$$\left(R(S_i)\right)_K \left(R(S_j)\right)_{n+K} + \left(R(S_i)\right)_{K+n} \left(R(S_j)\right)_K = 0 \mod(2) \iff [S_i, S_j] = 0$$

CASE BY CASE:

- If $S_i = \mathbb{1}$ or $S_j = \mathbb{1}$ $\Rightarrow$ ok
- If $S_i = X$ $\Rightarrow$ $S_j = X$ $\Rightarrow$ $1\cdot 0 + 0\cdot 1 = 0$ $\Rightarrow$ ok $[X, X] = 0$
  - $S_j = Y$ $\Rightarrow$ $1\cdot 1 + 0\cdot 1 = 1$ $\Rightarrow$ ok $\{X, Y\} = 0$
  - $S_j = Z$ $\Rightarrow$ $1\cdot 1 + 0\cdot 0 = 1$ $\Rightarrow$ ok $\{X, Z\} = 0$
- If $S_i = Z$ $\Rightarrow$ $S_j = Y$ $\Rightarrow$ 1 $\Rightarrow$ ok
  - $S_j = Z$ $\Rightarrow$ 1 $\Rightarrow$ ok
- If $S_i = Y$ $\Rightarrow$ $S_j = Y$ $\Rightarrow$ ok

TH (26)

A) If $S_1, \dots, S_n$ commute (easy to verify)

B) $S_1, \dots, S_n$ are check-independent (Gauss elimination of check-matrix)

C) $S_i^2 = \mathbb{1}$ $\forall S_i \in Gen(S)$ ($S_i = \pm P$ with $P \in \{\mathbb{1}, X, Y, Z\}^{\otimes n}$)

D) $S_i \neq -\mathbb{1}$ $\forall S_i \in Gen(S)$

$\Rightarrow \dim\left(V_{S=\langle S_1, \dots, S_n \rangle}\right) = 2^{n-k}$

- Check problem sheet 9 ex. 2.